

Sanzhar Kali | Denver, CO

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Skills

Programming: Python, C++, SQL, x86 Assembly

Hardware & RTL: Verilog, SystemVerilog, CMOS, VLSI, Digital Logic, Computer Architecture

Tools & Platforms: Cadence Virtuoso, LTspice, Linux, Git, Bash, Oscilloscopes, Signal Generators

Networking & Systems: Wireshark, Client-Server Models, Packet Analysis

Education

University of Illinois at Chicago

Chicago, IL

- *Bachelor of Science, Electrical & Computer Engineering* August 2021 - December 2025

Experience

Continental AG

Chicago, IL

Senior Design Project - (Hardware & Software Systems)

August 2024 - May 2025

- Designed a Python based GUI enabling non-technical users to configure and execute hardware test cases
- Implemented real time thermal data visualization and logging for system level analysis
- Developed a closed loop temperature control algorithm using sensor feedback and rate of change analysis
- Integrated hardware subsystems via structured CSV and sensor data pipelines to automate test execution

Projects

Software projects

Livestock monitoring

- Developed a Raspberry Pi, Arduino system with GPS based geofencing logic using Python
- Built a mobile accessible web interface for real time monitoring and configuration for fence using C++

Campus Navigation

- Modeled UIC campus as a weighted graph and implemented Dijkstra's algorithm in C++
- Built a CLI tool to parse map data and compute shortest paths with distance outputs

Automated Course Seat Monitoring & Registration System

- Developed a Python based Telegram bot to monitor university course seat availability in real time
- Implemented authenticated session handling to automate course registration upon seat availability
- Designed event driven logic to trigger enrollment actions and user notifications with minimal latency

Hardware projects

Artificial Neuron

- Designed transistor level CMOS circuits modeling neuron thresholding behavior
- Simulated and validated weighted signal integration using LTspice and waveform analysis

FSM & VLSI Design

- Designed FSMs using flip-flops and combinational logic using CircuitVerse
- Verified RTL functionality in Verilog and mapped designs to physical IC hardware

Custom 8-bit Processor Subsystem

- Designed core processor components including ALU, register blocks, and control logic at the transistor level
- Implemented and simulated CMOS logic using Cadence Virtuoso to validate functional correctness and timing behavior
- Analyzed signal integrity, propagation delay, and switching behavior through waveform simulations